



General Chemistry I

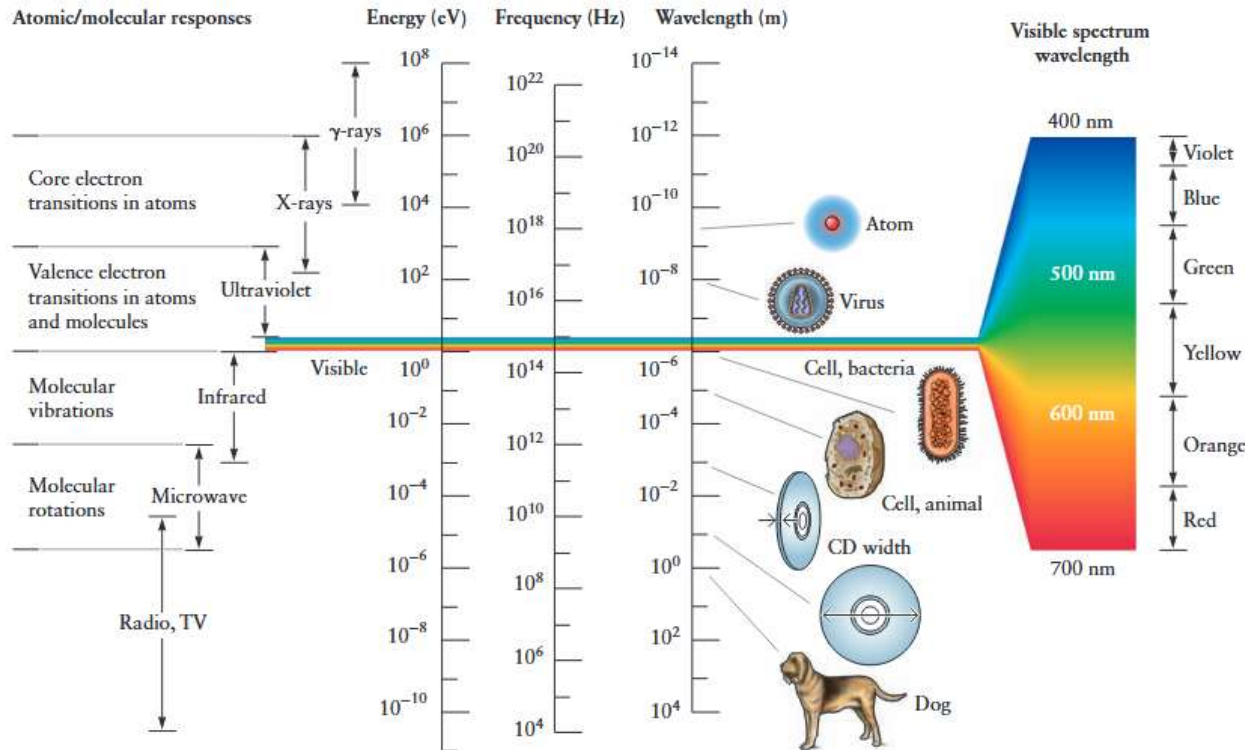
Recitation 4

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Quiz

1. Potassium atoms in a flame emit light as they undergo transitions from one energy level to another that is 4.9×10^{-19} J lower in energy. Calculate the wavelength of light emitted and, by referring to Figure 4.3, predict the color visible in the flame.



$$E = h\nu = h \frac{c}{\lambda}$$

$$\lambda = \frac{hc}{E} = \frac{h \cdot c}{4.9 \times 10^{-19} \text{ J}} = 4.06 \times 10^{-7} \text{ m} = 406 \text{ nm}$$

Violet (Purple?)

FIGURE 4.3 The electromagnetic spectrum. Note the small fraction that is visible to the human eye.

Quiz

2. He^+ ions are observed in stellar atmospheres. Use the Bohr model to calculate the radius and the energy of He^+ in the $n = 5$ state. How much energy would be required to remove the electrons from 1 mol of He^+ in this state? What frequency and wavelength of light would be emitted in a transition from the $n = 5$ to the $n = 3$ state of this ion? Express all results in SI units.

$$r = \frac{n^2}{Z} a_0 = 6.612 \times 10^{-10} m$$

$$E_n = -\frac{Z^2}{n^2} \quad (\text{rydberg}) \quad n = 1, 2, 3, \dots$$

$$1 \text{ rydberg} = 2.18 \times 10^{-18} \text{ J}$$

$$E_5 = \frac{4 * R_H}{25} = \frac{4 * 2.178 * 10^{-18}}{25} J = 3.48 * 10^{-19} J$$

to remove the electrons: $n=5$ to $n=\infty$

$$E_{total} = N_A E_5 = 2.098 * 10^{-10} J \cdot mol^{-1}$$

$n=5$ to $n=3$

$$\Delta E = E_5 - E_3 = -\frac{4 * R_H}{25} - \left(-\frac{4 * R_H}{9} \right) = 6.195 * 10^{-19} J$$

$$\lambda = \frac{hc}{\Delta E} = 321 nm$$

Homework 2

1. In a gaseous RbF molecule, the bond length is 2.274×10^{-10} m. Using data from Appendix F and making the oversimplified assumption as on the shape of the potential curve from $\text{Rb}^+ + \text{F}^-$ to an internuclear separation of 2.274×10^{-10} m, calculate the energy in kJ mol^{-1} required to dissociate RbF to neutral atoms.

Homework 2

2. Estimate the percent ionic character of the bond in each of the following species. All the species are unstable or reactive under ordinary laboratory conditions, but they can be observed in interstellar space.

	Bond Length (Å)	Dipole Moment (D)
OH	0.980	1.66
CH	1.131	1.46
CN	1.175	1.45
C ₂	1.246	0

$$\mu(\text{D}) = [R(\text{\AA})/0.2082 \text{\AA}] \delta$$

Homework 2

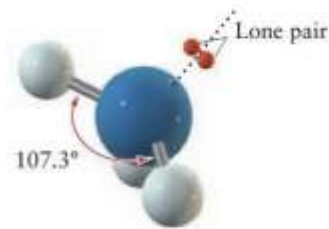
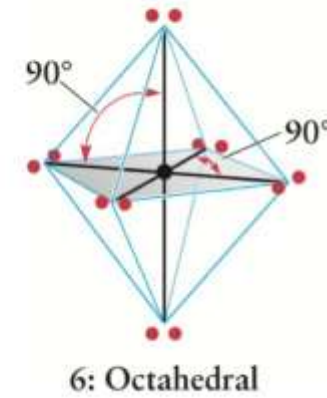
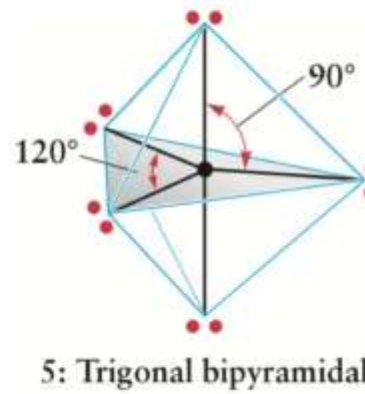
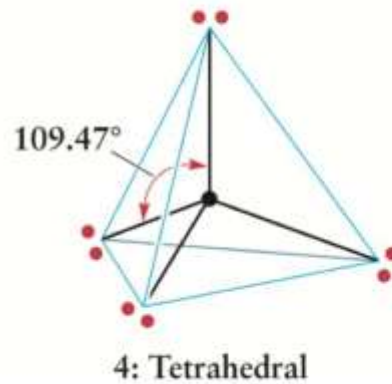
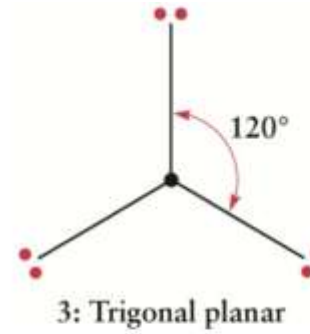
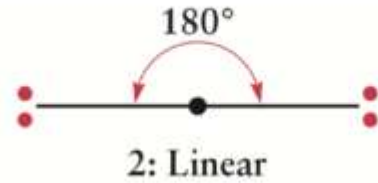
3. Draw Lewis electron dot diagrams for the following species, indicating formal charges and resonance diagrams where applicable.
- (a) HNC (central N atom)
 - (b) SCN^- (thiocyanate ion)
 - (c) H_2CNN (the first N atom is bonded to the carbon and the second N)
4. For each of the following molecules or molecular ions, give the steric number and sketch and name the approximate molecular geometry. In each case, the central atom is listed first and the other atoms are all bonded directly to it.
- (a) PF_3 (b) SO_2Cl_2 (c) PF_6^- (d) ClO_2^- (e) GeH_4

Homework 2

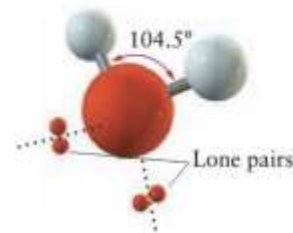
5. Ozone (O_3) has a nonzero dipole moment. In the molecule of O_3 , one of the oxygen atoms is directly bonded to the other two, which are not bonded to each other.
- (a) Based on this information, state which of the following structures are possible for the ozone molecule: symmetric linear, nonsymmetric linear (for example, different O-O bond lengths), and bent. (Note: Even an O-O bond can have a bond dipole if the two oxygen atoms are bonded to different atoms or if only one of the oxygen atoms is bonded to a third atom.)
- (b) Use the VSEPR theory to predict which of the structures of part (a) is observed.

Bonds and molecular structure

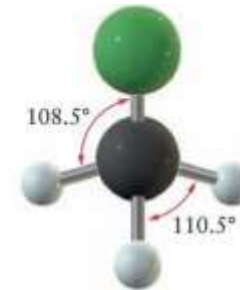
SN= bond pair + lone pair



NH_3



H_2O



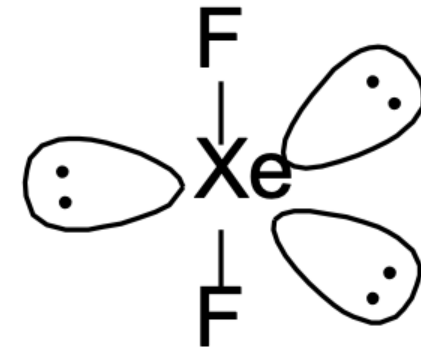
CH_3Cl

Bonds and molecular structure

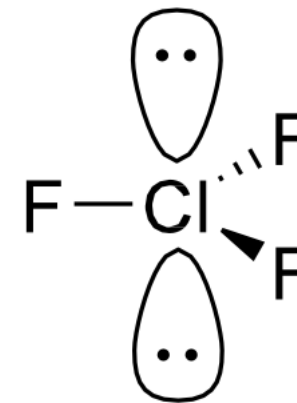
SN= Total amount of orbitals

SN of A	Molecular type X is a bonded atom; E is a lone pair	Predicted shape
2	AX ₂	Linear (线形)
3	AX ₃	Trigonal planar (三角形)
	AX ₂ E	Bent (弯曲形)
4	AX ₄	Tetrahedral (四面体)
	AX ₃ E	Trigonal pyramidal (三角锥)
	AX ₂ E ₂	Bent
5	AX ₅	Trigonal bipyramid (三角双锥)
	AX ₄ E	Distorted see-saw (跷跷板)
	AX ₃ E ₂	Distorted T (T字)
	AX ₂ E ₃	Linear
6	AX ₆	Octahedral (八面体)
	AX ₅ E	Square pyramidal (四角锥)
	AX ₄ E ₂	Square planar (正方平面)

SN=3 $\longrightarrow$ $sp^2(1 + 2 = 3)$

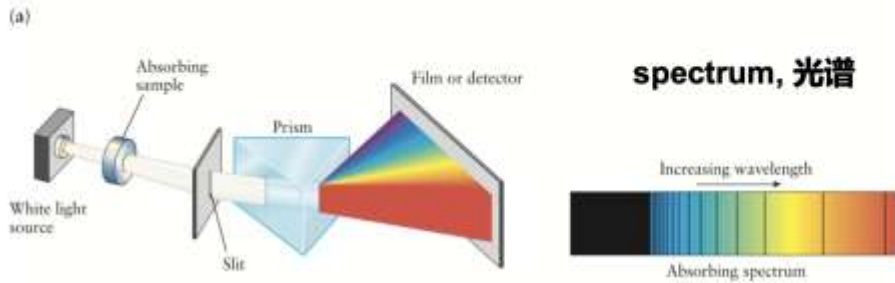
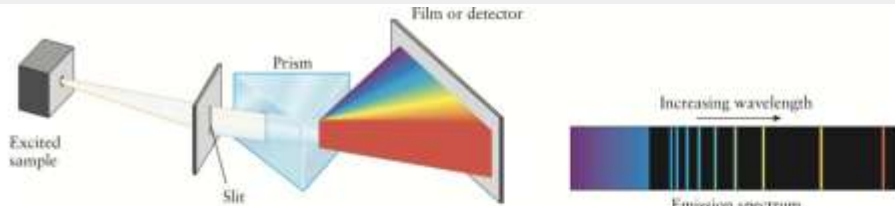


Why linear?



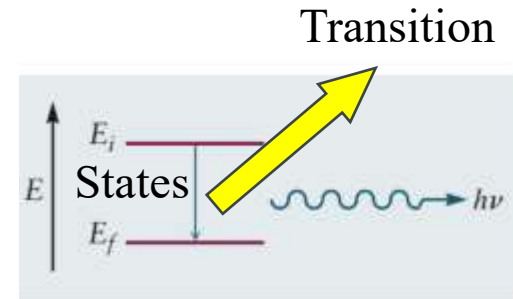
Why planar triangle?

Quantum mechanics and atom structure



spectrum, 光谱

Emission spectrum

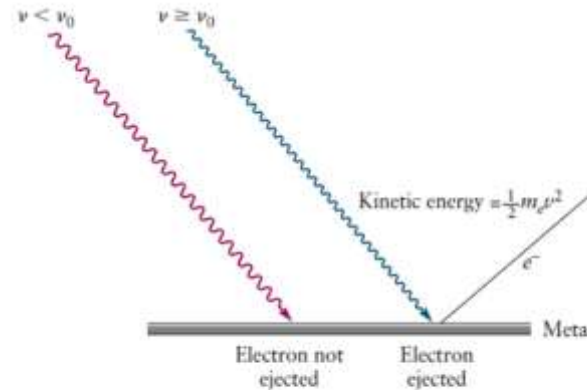


$$\nu = \frac{E_f - E_i}{h} \quad \text{or} \quad \Delta E = h\nu$$

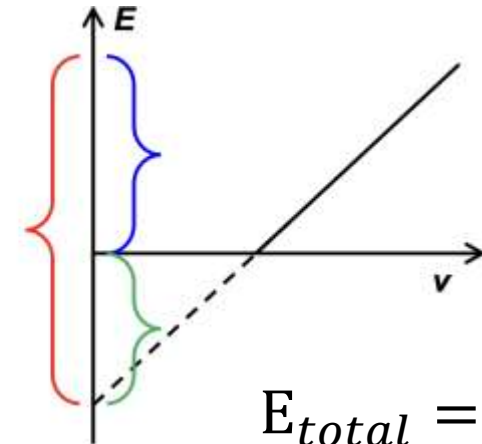
Photon(light)

h is Planck's constant

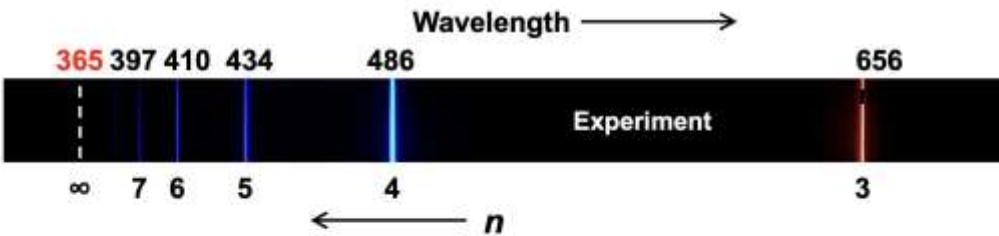
Absorption spectrum



$$E_{\max} = \frac{1}{2} m v_e^2 = h\nu - \phi$$



$$E_{\text{total}} = E_k + E_p$$



$$\frac{656}{365} \approx 1.80 = \frac{9}{5} = \frac{9}{9-4}$$

$$\frac{486}{365} \approx 1.33 = \frac{4}{3} = \frac{16}{16-4}$$

$$\Rightarrow \lambda_n = B \frac{n^2}{n^2 - 4}, \quad B = 365 \text{ nm}, \quad n = 3, 4, 5, \dots$$

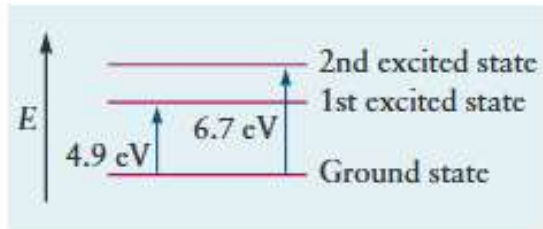
$$\nu = \left[\frac{1}{4} - \frac{1}{n^2} \right] \times 3.29 \times 10^{15} \text{ s}^{-1} \quad n = 3, 4, 5, \dots$$

1905: Albert Einstein's Year of Miracles

$$E_{k,\max} + \phi = E_{\text{light}} \propto \nu$$

$$E_{\text{light}} = h\nu \quad \text{Planck's constant}$$

Quantum mechanics and atom structure



Simplified energy-level diagram for mercury.

EXAMPLE 4.2

The first two excitation voltage thresholds in the Franck–Hertz study of mercury vapor were found at 4.9 and 6.7 V. Calculate the wavelength of light emitted by mercury atoms after excitation past each of these thresholds.

Solution

The emitted wavelength at a particular value of V_{thr} is given by the following equation:

$$\begin{aligned}\lambda &= \frac{hc}{\Delta E} = \frac{hc}{eV_{\text{thr}}} = \frac{(6.6261 \times 10^{-34} \text{ J s})(2.9979 \times 10^8 \text{ ms}^{-1})}{(1.6022 \times 10^{-19} \text{ C})(V_{\text{thr}}[\text{V}])} \\ &= \frac{1239.8 \text{ nm}}{V_{\text{thr}}[\text{V}]}\end{aligned}$$

The value of each emission wavelength is calculated by substituting in a particular value of V_{thr} expressed in units of volts (V).

At $V_{\text{thr}} = 4.9 \text{ V}$, the calculated wavelength is $\lambda = 250 \text{ nm}$. The wavelength actually observed above this threshold was 253.7 nm.

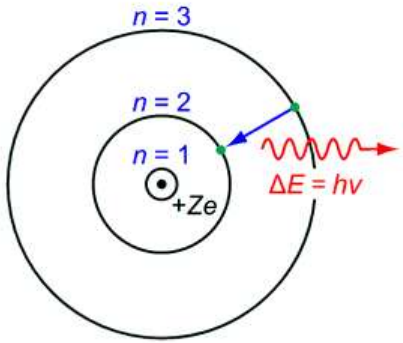
At $V_{\text{thr}} = 6.7 \text{ V}$, the calculated wavelength is $\lambda = 180 \text{ nm}$. The wavelength actually observed above this threshold was 184.9 nm.

Energy differences measured by the Franck–Hertz method and by optical emission spectroscopy agree quite closely. The optical measurements are more precise. These results enable us to begin to construct the energy diagram for mercury, showing the location of the first two excited states relative to the ground state.

Related Problems: 17, 18

Quantum mechanics and atom structure

Bohr Model



Bohr model:

- only certain discrete orbits are allowed (characterized by radius r_n and energy E_n)
- light is emitted or absorbed only when the electron "jumps" from one stable orbit to another

"light is emitted or absorbed only when the electron 'jumps' from one stable orbit to another."

The total energy of the hydrogen atom:

$$E = \frac{1}{2} m_e v^2 - \frac{Z e^2}{4 \pi \epsilon_0 r}$$

Kinetics

potential

Find it in your own calculator

The angular momentum of the electron is quantized:

$$L = m_e v r = n \left(\frac{h}{2\pi} \right) \quad n = 1, 2, 3, \dots$$

$$\frac{Z e^2}{4 \pi \epsilon_0 r^2} = m_e \frac{v^2}{r}$$



$$r_n = \frac{\epsilon_0 n^2 h^2}{\pi Z e^2 m_e} = \frac{n^2}{Z} a_0$$

The existence of discrete orbits and quantized energies follows directly as a consequence of the quantization of angular momentum

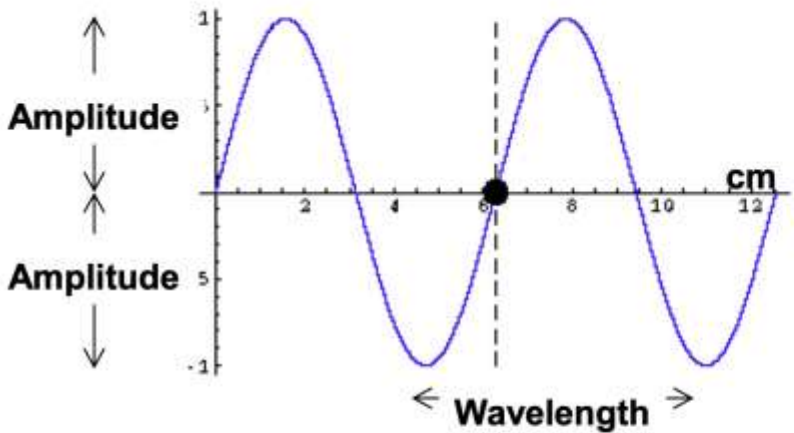
$$E_n = -\frac{Z^2}{n^2} \quad (\text{rydberg}) \quad n = 1, 2, 3, \dots$$

$$\Delta E = h\nu = \frac{Z^2 e^4 m_e}{8 \epsilon_0^2 h^2} \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right] \quad (\text{emission})$$

Wave and wave functions

$$E = h\nu = h \frac{c}{\lambda}$$

$$\lambda = \frac{c}{\nu}$$



sine wave 正弦波

Phase (φ) propagates in both time and space.

$$\varphi(x) = \frac{2\pi x}{\lambda} = 2\pi \tilde{\nu} x$$

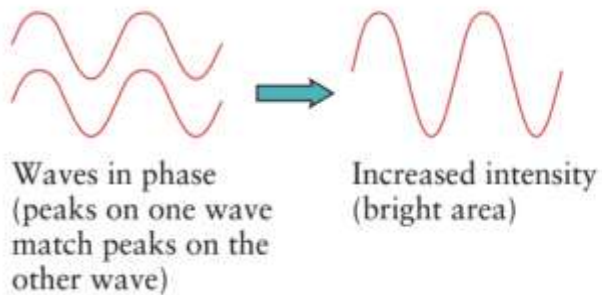
$$\varphi(t) = -\frac{2\pi t}{T} = -2\pi \nu t$$

$$y(x,t) = A \sin [\varphi(x) + \varphi(t)] = A \sin \left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T} \right) \right]$$

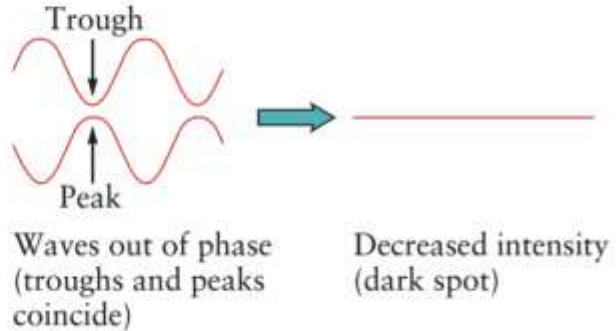
Interference is a vital property for wave, as well as the wave functions.

Amplitude 振幅	$A = 1$
Wavelength 波长	$\lambda = 6.3 \text{ cm} \leftrightarrow$ Wavenumber 波数 $\tilde{\nu} = \lambda^{-1} \approx 0.16 \text{ cm}^{-1}$
Period 周期	$T = 3.0 \text{ s} \leftrightarrow$ Frequency 频率 $\nu = T^{-1} \approx 0.33 \text{ Hz}$
Speed 波速	$s = \lambda / T = \lambda \nu \approx 2.1 \text{ cm} \cdot \text{s}^{-1}$

(a) Constructive interference



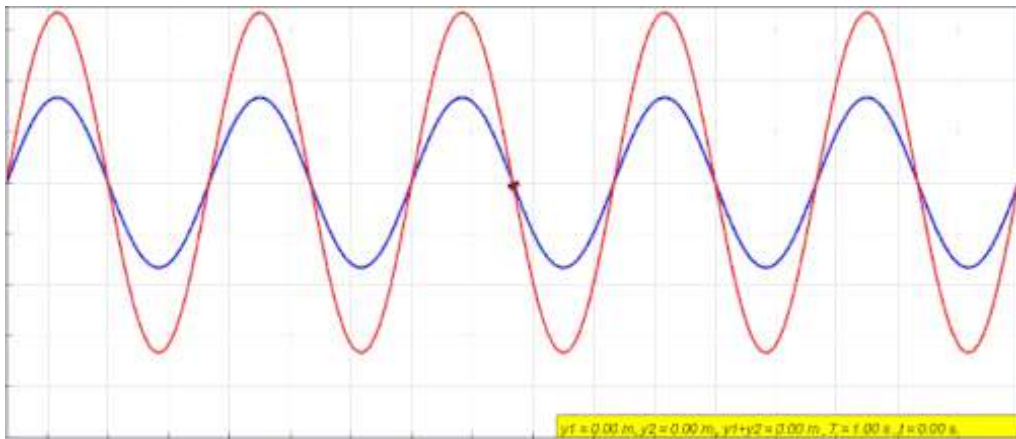
(b) Destructive interference



Some important parameters

Wave and wave functions

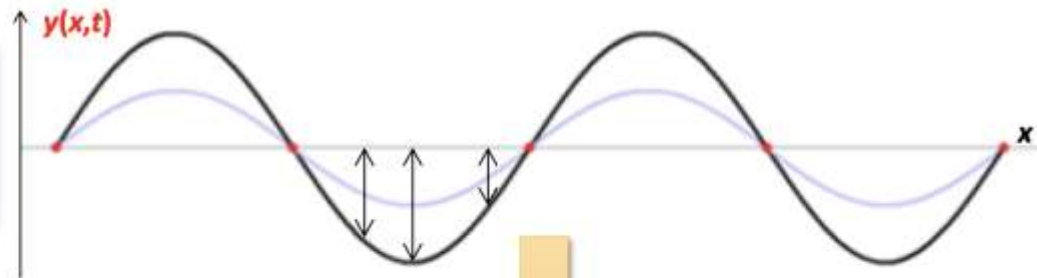
Standing wave: Equals the sum of two traveling waves in opposite directions



Animation of a standing wave (red) created by the superposition of a left traveling (blue) and right traveling (green) wave

$$y(x,t) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right) \cos\left(2\pi \frac{t}{T}\right)$$

波函数



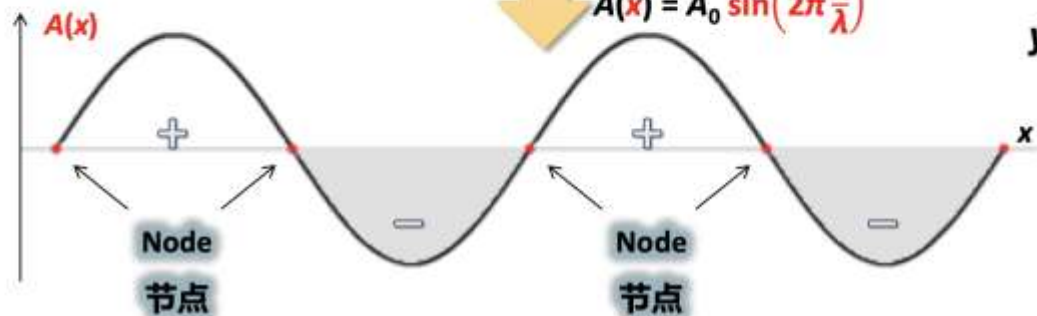
Amplitudes

$$A(x) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right)$$

$$y(x,t) = \frac{A_0}{2} \sin\left[2\pi \left(\frac{x}{\lambda} - \frac{t}{T}\right)\right] + \frac{A_0}{2} \sin\left[2\pi \left(\frac{x}{\lambda} + \frac{t}{T}\right)\right]$$

$$y(x,t) = A_0 \sin\left(2\pi \frac{x}{\lambda}\right)$$

Stationary State 定态

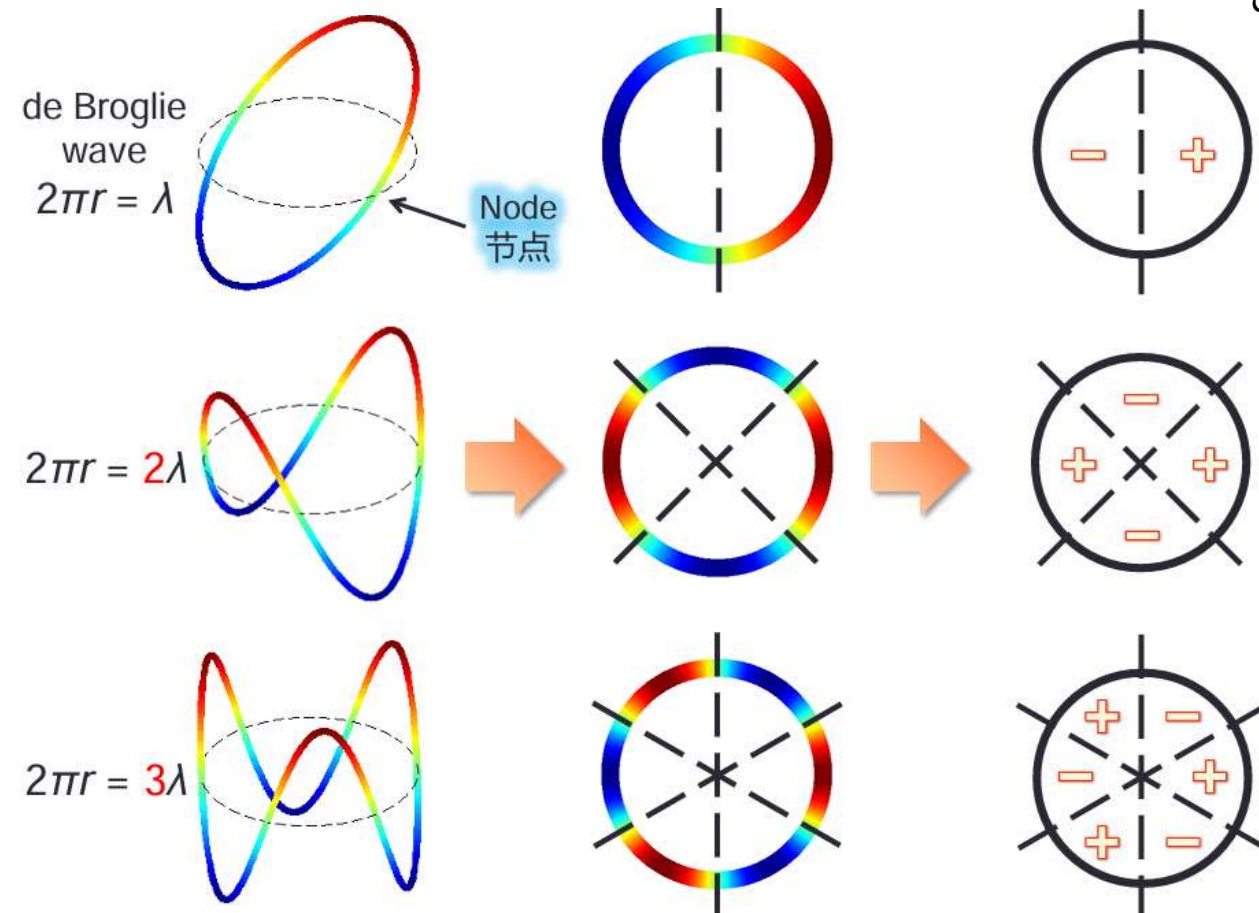


Wave and wave functions

“electron might be associated with a standing wave”

De Broglie wave (Matter wave)

$$\lambda = \frac{h}{m_e v} = \frac{h}{p}$$



An integral number of wavelengths must fit into the circumference of the circle

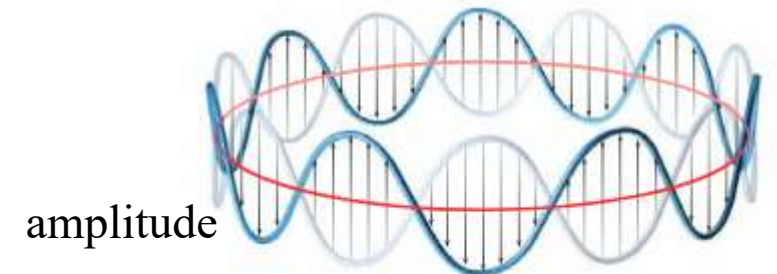
$$n \frac{\lambda}{2} = L \quad n = 1, 2, 3, \dots$$

$$n\lambda = 2\pi r \quad n = 1, 2, 3, \dots$$

From Bohr's model

$$m_e v r = n \frac{h}{2\pi}$$

$$2\pi r = n \left[\frac{h}{m_e v} \right]$$



How many nodes can we obtain?

Wave and wave functions

Calculate the de Broglie wavelengths of (a) an electron moving with velocity $1.0 \times 10^6 \text{ m s}^{-1}$ and (b) a baseball of mass 0.145 kg , thrown with a velocity of 30 m s^{-1} .

Solution

$$\begin{aligned} \text{(a)} \quad \lambda &= \frac{h}{p} = \frac{h}{m_e v} = \frac{6.626 \times 10^{-34} \text{ J s}}{(9.11 \times 10^{-31} \text{ kg})(1.0 \times 10^6 \text{ m s}^{-1})} \\ &= 7.3 \times 10^{-10} \text{ m} = 7.3 \text{ \AA} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \lambda &= \frac{h}{mv} = \frac{6.626 \times 10^{-34} \text{ J s}}{(0.145 \text{ kg})(30 \text{ m s}^{-1})} \\ &= 1.5 \times 10^{-34} \text{ m} = 1.5 \times 10^{-24} \text{ \AA} \end{aligned}$$

The latter wavelength is far too small to be observed. For this reason, we do not recognize the wavelike properties of baseballs or other macroscopic objects, even though they are always present. However, on a microscopic level, electrons moving in atoms show wavelike properties that are essential for explaining atomic structure.

Wave and wave functions

The Heisenberg Principle (海森堡不确定性原理)

Suppose photons of green light (wavelength 5.3×10^{-7} m) are used to locate the position of the baseball from Example 4.5 with precision of one wavelength. Calculate the minimum spread in the *speed* of the baseball.

Solution

The Heisenberg relation,

$$(\Delta x)(\Delta p) \geq h/4\pi$$

gives

$$\Delta p \geq \frac{h}{4\pi\Delta x} = \frac{6.626 \times 10^{-34} \text{ J s}}{4\pi(5.3 \times 10^{-7} \text{ m})} = 9.9 \times 10^{-29} \text{ kg m s}^{-1}$$

Because the momentum is just the mass (a constant) times the speed, the spread in the speed is

$$\Delta v = \frac{\Delta p}{m} \geq \frac{9.9 \times 10^{-29} \text{ kg m s}^{-1}}{(0.145 \text{ kg})} = 6.8 \times 10^{-28} \text{ m s}^{-1}$$

This is such a tiny fraction of the speed of the baseball (30 m s^{-1}) that indeterminacy plays a negligible role in the measurement of the baseball's motion. Such is the case for all macroscopic objects.

$$(\Delta x)(\Delta p) \geq h/4\pi$$

Wave and wave functions

Probability and wave function

Intensity of the light wave (can be measured) is proportional to the square of the amplitude of the electric field: $\text{intensity} \propto (E_{\text{max}})^2$

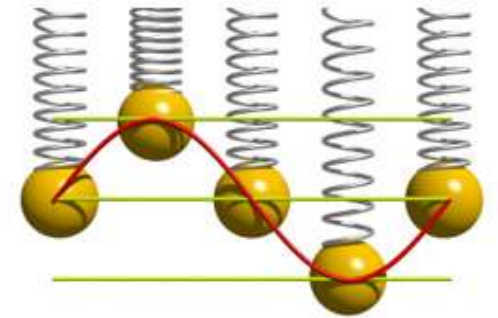
If we view electromagnetic radiation as a collection of particles (photons), then the intensity is simply proportional to the density of photons in a region of space. Connecting the wave and particle views of the electromagnetic field suggests that the probability of finding a photon is given by the square of the amplitude of the electric field

In this case, we consider ψ^2 as probability density for that particle

For a simple harmonic motion $E = \frac{k}{2} A^2$

Let wave function $\psi(x) = \sqrt{\frac{k}{2}} A(x)$

then $E = [\psi(x)]^2$



$$\int_{-\infty}^{+\infty} P(x) dx = \int_{-\infty}^{+\infty} [\psi(x)]^2 dx = 1$$

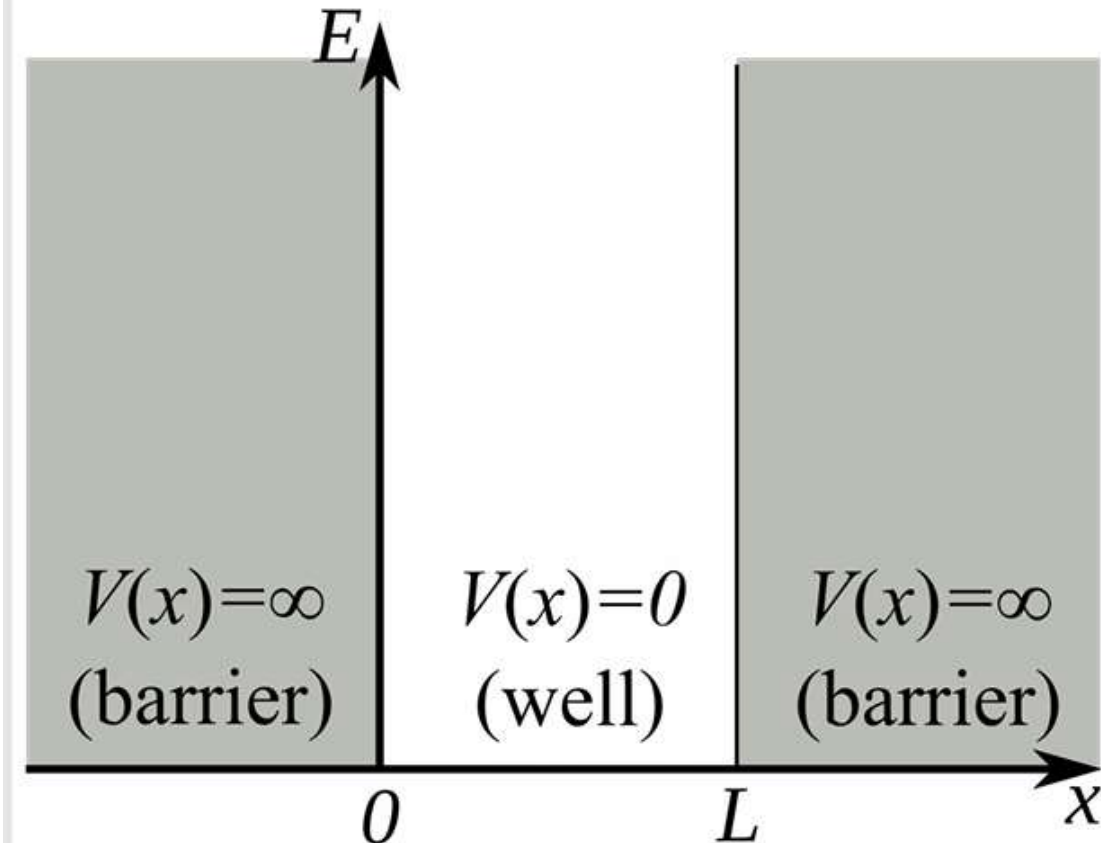
$P(x)$: the probability density in x position

“The sum of every value of possibility is 1(100%)”

Introduction of quantum mechanics

Particle-in-a-Box models: a particle confined by potential energy barriers to a certain region of space.

1D box: the particle is located on the x-axis in the interval between 0 and L (L is the length of the box)



- The potential energy $V(x)$ rise abruptly to an infinite value at the two end walls to prevent fast-moving particles from escaping.
- Inside the box, the motion of the particle is free, so $V(x) = 0$

$$V(x) = \begin{cases} 0, & x_c - \frac{L}{2} < x < x_c + \frac{L}{2}, \\ \infty, & \text{otherwise,} \end{cases},$$

$$\psi(x) = 0 \text{ for } x \leq 0 \text{ and } x \geq L$$

Introduction of quantum mechanics

Schrödinger Equation for a particle in a box:

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Inside the box, where $V = 0$, the Schrödinger equation has the following form

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

or equivalently,

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2mE}{h^2} \psi(x)$$

a potential solution?

$$\psi(x) = A \sin kx$$

正余弦函数本身是其两阶负导数

The boundary conditions require: $\psi(x) = 0$ at $x = 0$  • The cosine function can be eliminated, $\cos(0) = 1$
• The sine function meets the requirement, $\sin(0) = 0$

The boundary conditions require that $\psi(x) = 0$ at $x = L$  $\psi(L) = 0$
 $\psi(L) = A \sin kL = 0$
 $kL = n\pi \quad n = 1, 2, 3, \dots$

$$\psi(x) = A \sin \left(\frac{n\pi x}{L} \right) \quad n = 1, 2, 3, \dots$$

The wave function must be normalized $A^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$

Don't forget we just talked about probability!



$$A^2 \left(\frac{L}{2} \right) = 1 \quad A = \sqrt{\frac{2}{L}}$$

The normalized wave function for the one-dimensional particle in a box is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \quad n = 1, 2, 3, \dots$$

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

$$\frac{d^2\psi(x)}{dx^2} = -\frac{8\pi^2mE}{h^2} \psi(x)$$

$$\begin{aligned} \frac{d^2\psi_n(x)}{dx^2} &= \frac{d^2}{dx^2} \left[\sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \right] \\ &= -\left(\frac{n\pi}{L} \right)^2 \left[\sqrt{\frac{2}{L}} \sin \left(\frac{n\pi x}{L} \right) \right] \\ &= -\left(\frac{n\pi}{L} \right)^2 \psi_n(x) \end{aligned}$$



$$\frac{8\pi^2mE_n}{h^2} = \frac{n^2\pi^2}{L^2}$$

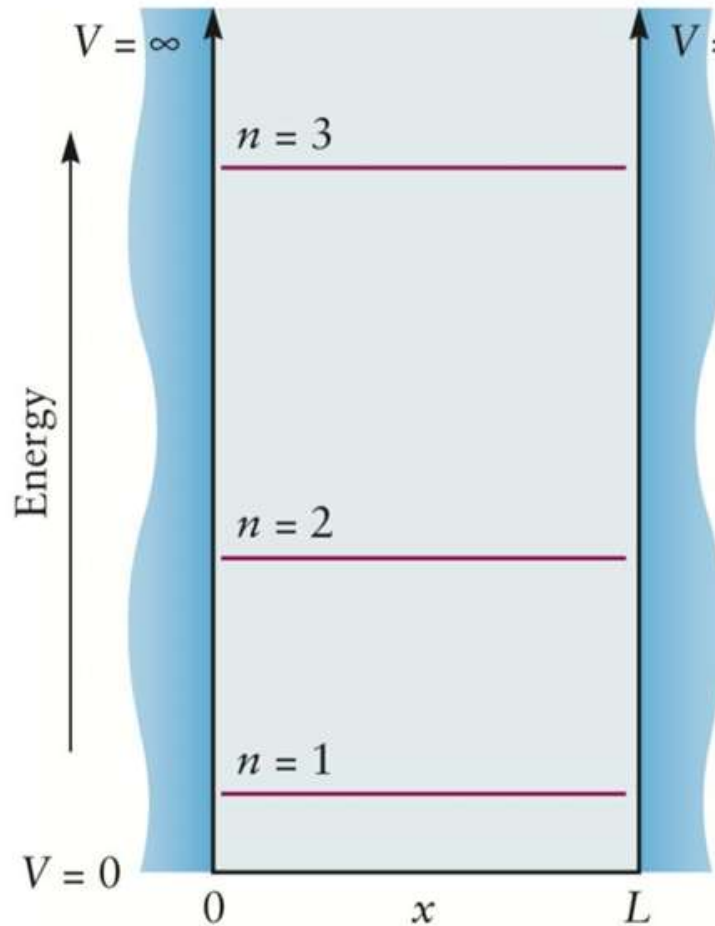
$$E_n = \frac{n^2h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$

The **energy** of a particle in the box is quantized

Introduction of quantum mechanics

Particle-in-a-Box models

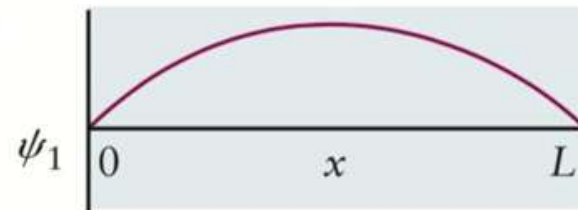
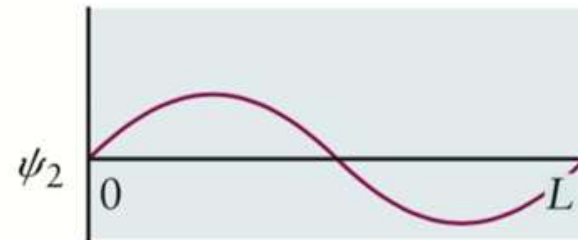
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \dots$$



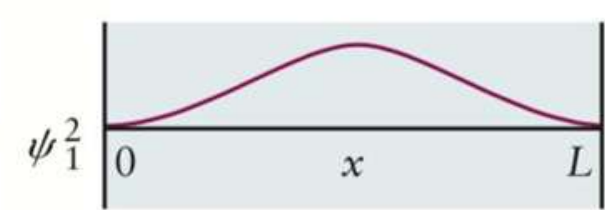
(a) Energy levels



Wherever there is a node in the wave function, the probability is zero that the particle will be found at that location.



(b) Wave functions



(c) Probabilities

The wave function ψ_n has $n - 1$ nodes